# Jiajin (David) Wu

## AI/LLM Software & Application Engineer

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### Summary

AI/LLM software and application engineer with 3+ years building production Python/TypeScript systems for enterprise workflows. At Aptiv, delivered 3 workflow agents and a multi-agent platform with LangGraph, hybrid retrieval-augmented generation (RAG), human review controls, FastAPI, and a React/Next.js console. Led the handling of 430+ issues across 9 L1–L3 ADAS programs and earned a 121% top-tier performance rating for two consecutive years.

### Technical Skills

**AI/LLM Engineering:** LangGraph, LangChain, deepagents, Agno, MCP, GitHub Copilot SDK, RAG, tool calling, LLM evaluation, OpenAI, Anthropic Claude, Google Gemini, human-in-the-loop
**Languages & Frameworks:** Python, TypeScript, React, Next.js, FastAPI, Pydantic, SQLAlchemy, SQL, MATLAB
**Data & Retrieval:** OpenSearch, BM25, embeddings, kNN/vector search, hybrid search, SQLite, MinIO/S3, semantic memory
**Platform & Integration:** REST APIs, asyncio/aiohttp, JWT/RBAC/OAuth, Jenkins CI/CD, Docker, Microsoft Graph, Jira, Polarion, Confluence, pytest, Ruff
**Systems Engineering:** ADAS L1–L3, CAN/CANoe, MBSE, SysML, MagicDraw/MagicGrid, requirements, verification, root-cause analysis

### Work Experience

**Vehicle System Triage Engineer — Technical Lead, VSDA Triage Team** **Feb 2023 — Present**
*Aptiv Corporation* *Troy, MI*

- Built and deployed 3 Python AI/LLM agents: Release Email Analyzer extracts software versions/change sets and triggers stream-definition builds; VSDA Email Agent converts VSI/VSDA reports into release reports; Jira Ticket Extractor populates the triage database
- Engineered a Jira REST API ticket-automation tool that generated 1,000+ standardized stories team-wide; automated work-history reporting with Jira APIs and Jenkins for near-real-time management visibility
- Led cross-functional issue triage across 9 L1–L3 ADAS programs, personally triaging 267 and directing 430+ system-, vehicle-, and bench-level tickets across Algorithm, Integration, Testing, and Data-Mining teams
- Built reusable Python/MATLAB batch pipelines for large AEB/PEB datasets, calculating camera/module latency and comparing AHBC performance before/after coating changes to deliver customer-ready analysis
- Automated Polarion test-plan generation, test-case import, and result updates; authored 38 integration and radar-signal test cases with end-to-end requirements traceability
- Earned a 121% top-tier performance rating with “Exceeds Expectations” across all goals for two consecutive years; represented the team in customer meetings and company-wide AI working sessions

**Graduate Student Instructor — Systems Engineering (ISD 521)** **Aug — Dec 2022**
*University of Michigan, College of Engineering* *Ann Arbor, MI*

- Co-guided 51 students across 13 teams through a four-deliverable MBSE project worth 40% of the course grade, translating stakeholder needs into SysML context, behavior, traceability, architecture, and validation models in MagicDraw/MagicGrid
- Authored a five-slide technology-readiness recitation and an 83-formula Excel decision-tree model comparing 3 investment strategies through expected-payoff and value-of-information analysis

**Systems Engineering Intern** **Jun — Sept 2022**
*Varian Medical System, a Siemens Healthiness Company* *Palo Alto, CA*

- Designed a MATLAB GUI application that generated parameterized XML test plans, streamlining execution of radiotherapy-system verification workflows
- Built MATLAB pipelines to acquire, process, and visualize beam data for control-system diagnosis and Beam Generation Monitor subsystem verification
- Collaborated with mechanical, electrical, and software engineers to test a next-generation radiotherapy treatment-system prototype and document results

## Additional Work Experience

**Mechanical Engineering Co-op** **May — Aug 2021**
*Contemporary Intelligent Manufacturing Co. Ltd.* *Ningde, China*

- Led development and production-line deployment of a portable, non-contact electrode-thickness measurement system meeting a $10,000 maximum-cost requirement and a 0.2-micrometer resolution target

**Systems Engineering Intern** **Mar — Jun 2020**
*Shanghai MAXIEYE Technology Co. Ltd.* *Shanghai, China*

- Developed L2 collision-warning requirements and verification plans, then wrote MATLAB tooling to clean and visualize CAN signals for system validation
- Analyzed NHTSA human-factors guidance for L2+/L3 ADAS development and translated findings into system-level engineering inputs

## Selected AI/LLM Projects

**VSDA Deep Agent — Full-Stack Enterprise AI Platform** **Python, LangGraph, FastAPI, OpenSearch**

- Built a LangGraph multi-agent platform with OpenSearch hybrid RAG, curated wiki retrieval, isolated code analysis, and access controls for Jira/Polarion/Confluence tools. Tied write approvals to payload files verified with SHA-256
- Developed a React/Next.js console to stream model output and run state, track cost budgets, and manage scheduled workflows. Added execution history, checkpoint recovery, and approvals tied to task checkpoints

**Resume Tailor Harness — Fact-Locked AI Job Pipeline** **Python, FastAPI, React, TypeScript**

- Added deterministic fact checks and configurable reviewer panels to keep generated resumes and cover letters grounded in source material. Implemented 8 Typst layouts, automatic fitting within page limits, and visual review of rendered PDFs
- Built application APIs that isolate tenant data, reuse profile answers, and check revisions. Connected Gmail, Outlook, and IMAP to a bilingual React/FastAPI app so users can review suggested status changes before applying them

**Requirement Buddy — Reviewed Engineering Workflows** **Python, MCP, GitHub Copilot SDK**

- Built resumable Copilot SDK workflows and a dashboard for writing and independently reviewing requirements, architecture, and test cases. Tracked artifact sources and kept validated records of workflow responses
- Implemented 3 MCP connectors for Jira/Polarion/Confluence with source-revision checks and authorized publishing. Verified writes by reading results back; added Polarion test-run, execution-record, and step-result management

**Enterprise Engineering Automation Suite** **Python, REST APIs, Jenkins, Jinja2**

- Built typed async clients for Jira, Polarion, Confluence, and Microsoft Graph with bounded hierarchy queries, metadata-aware authoring, and version checks. Added read retries while keeping each write to a single attempt
- Made Polarion PDF exports recover from dropped connections and expired sessions, with a deadline for each document and ordered pagination. Maintained adaptive bulk scheduling and Jenkins reports that resume from checkpoints

## Education

**Master of Engineering, Systems Engineering & Design** — University of Michigan–Ann Arbor Dec 2022, GPA: 3.9
**Bachelor of Science, Mechanical Engineering** — Shanghai Jiao Tong University Aug 2021, GPA: 3.6

## Publications

M. Liu, F. Zhao, **J. Wu**, W. Fan, J. Zhang, D. Hung, *Hybridizing Unsupervised Clustering Methods for In-cylinder Vortex Motion Analysis Under Different Swirl Ratio Conditions*, SAE WCX, 2021

**J. Wu**, J. Wang, Y. Yang, Z. Zeng, D. Lu, and L. Lian, *Hybrid Aerial-Aquatic Vehicle for Large Scale High Spatial Resolution Marine Observation*, OCEANS, 2019